



THE MAIN TITLE OF THE THESIS

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YOUR NAME

THESIS SUBMITTED IN PARTIAL FULFILLMENT
OF THE REQUIREMENTS FOR THE DEGREE OF
BACHELOR OF SCIENCE IN COGNITIVE SCIENCE & ARTIFICIAL INTELLIGENCE

DEPARTMENT OF
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SCHOOL OF HUMANITIES AND DIGITAL SCIENCES
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WORD COUNT

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Some room for acknowledgements.

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Abstract

This is where the abstract goes. Don't forget to change the variables in `main.tex` to change all general placeholders shown in this document. The `frontmatter.tex` file should be left alone. The abstract should summarize in 150-250 words the contents of the thesis, mentioning: 1) the research goal, 2) the methods employed, 3) data collected or dataset/s used, 4) the main findings.

1 DATA SOURCE, ETHICS, CODE, AND TECHNOLOGY STATEMENT

Checklist for creating a Data Source, Ethics, Code, and Technology (DSECT) statement: Please provide answers to all below questions for transparency purposes when writing your ethical statement. There is no one-fits-all solution to ethical statements, and they heavily depend on the nature of the study. After the questions below, an example DSECT statement is provided.

- Data Source
 - o Who is the owner of the data?
 - o Does the thesis project involve collecting data from human participants or animals?
 - o Does the owner of the data give consent?
 - o How and through what channel is the data acquired?
- Figures
 - o Did you create all the images and figures?
 - o Do you have consent for the images that you did not generate?
- Code
 - o Did you use parts of the code from another study/someone else?

- o Did you list, including version number, all libraries and frameworks used?
- Technology
 - o Did you use any tools or services to paraphrase the given text (for example a thesaurus or the Academic Phrasebank)? Please name them.
 - o Did you use any tools or services to check spelling or grammar? Please name them.
 - o Did you use any tools or service to typeset the given text? Please name them.
 - o Did you use any reference management software other than the latex template? If so, please name it.
 - o The following **statement should be included**: During the preparation of this work, the author used [NAME TOOL / SERVICE] in order to [REASON]. After using this tool/service, the author reviewed and edited the content as needed and takes full responsibility for the content of the thesis.

A DSECT Statement can be generated by answering the questions listed above and excluding the questions when not applicable.

1.1 Source/Code/Ethics/Technology Statement Example

Data Source: The (DATA) has been acquired from the (DATA SOURCE) through an online request. The obtained data is anonymised. Work on this thesis did not involve collecting data from human participants or animals. The original owner of the data and code used in this thesis retains ownership of the data and code during and after the completion of this thesis. However, the institution was informed about the use of this data for this thesis and potential research publications. All the figures but Figure A and B belong to the author. To use Figure A and Figure B, the author received written permission via an email from FIGURE OWNER (AFFILIATION). The thesis code can be accessed through the GitHub repository following the link [CODE LINK]. Part of the CODE has been adapted by the author from (SOURCE, licensed under a CCo BY 4.0). The reused/adapted code fragments are clearly indicated in the notebook. In terms of writing, the author used assistance with the language of the paper. A generative language model (MODEL NAME + SOURCE) was used to improve the author's original content, for paraphrasing, spell checking and grammar. No other typesetting tools or services were used.

2 INTRODUCTION

Here you write the introduction. Start with briefly describing the research domain, why the topic is relevant, and current state-of-the-art by referencing pertinent research articles. Then, state your specific research goals/questions (you can separate them as shown below), followed by a one-paragraph summary of your findings. Discuss how answering your research questions can produce scientific relevance (domain insights) and societal relevance (real-world applications and/or benefits).

You can refer to chapters like the current Section 2. If you want to list research questions it's probably best to use quote and/or itemize, like so:

To what extent we define a main research question?

The sub-questions can be listed separately, as such:

RQ1 *How does sub question one influence the main RQ?*

RQ2 *To what degree does sub question add anything?*

Or, format it as you desire (tip: you can nest itemize as well). You can alternate *emph* and **textbf** however you wish. This should cover most of the things required for the introduction.

3 RELATED WORK

Here you provide a synthesis of the existing literature, comparing and drawing parallels across studies to establish the state-of-the-art. Cite relevant theories, models, and recent work. Describe what prior research has found, any contradicting findings or competing approaches. Identify any gaps or limitations in existing research and how your thesis potentially addresses these gaps in terms of methodology, data used, etc.

Copy paste BibTeX code¹ and put it in references.bib. After, you can cite some work (MacKay & Mac Kay, 2003) – using \citep. You can refer to the author of e.g. Minsky (1961) directly like using \cite (this does not work when using bracket-citation). If you use bracket-style,² you might want use \citeauthor when citing, like: see Ananny and Crawford Ananny and Crawford (2018). If you want to add pages you can use brackets in \citep[[p. 5]{mackay2003information}, which looks like: (MacKay & Mac Kay, 2003, p. 5). The first brackets can be used for things like *see*, and *e.g.*. If you want to cite multiple authors, simply comma-separate them (\citep{minsky1961steps,mackay2003information}) and

¹ Using e.g. the quote icon in GScholar, then BibTeX at the bottom.

² Find the natbib part in the main.tex L^AT_EX script.

it will aggregate them automatically (MacKay & Mac Kay, 2003; Minsky, 1961).

4 METHODS

The following sections describe the dataset, the extraction of physiological and aircraft features in sliding time windows, and the construction of multi-output targets. They also provide a brief summary of the machine learning model training and the metrics used for evaluation.

4.1 Dataset

The CogPilot dataset (?) was used in this project and is publicly available via PhysioNet (?). The dataset contains multimodal physiological, task performance, and interaction data collected from 35 participants who each performed flight-landing tasks at various difficulty levels. In a virtual reality environment, each participant followed an Instrument Landing System (ILS) approach and primarily relied on cockpit instrument readings to land an aircraft along a prescribed ideal flight path. Each experimental session consisted of three blocks with four missions per block, resulting in twelve missions per participant, with one difficulty level per mission.

Multiple wearable sensors were used to record seven physiological modalities, with nominal sampling rates ranging from 128 to 1024 Hz. These modalities comprise electromyography (EMG), accelerometry (ACC), electrodermal activity (EDA), electrocardiography (ECG), respiration (RESP), and eye tracking (ETK). A detailed overview of the functional data derived from each physiological signal is provided in Table 4 in Appendix A. Mission difficulty was manipulated via four distinct weather conditions. Flight performance measures were derived from the simulated aircraft behavior and grouped into three components. Two components represent the deviation from the ideal vertical and horizontal paths leading to the runway, namely the glideslope and localizer errors, both ranging from -3 to 3 degrees. The third component is the absolute deviation from the ideal airspeed of 115 knots, where the extreme airspeed error was mapped to a -3 to 3 scale by dividing the raw error (in knots) by 10.

4.2 Feature Engineering

Recordings of ECG and RESP were selected as the physiological modalities for feature preprocessing and extraction. This choice was informed by previous work indicating that heart rate (HR) derived from ECG can be

used as a practical signal for real-time adaptation of task difficulty to maintain trainees within an optimal workload range (?), and that respiratory behaviour is a robust marker of cognitive processing, with mentally demanding tasks typically associated with an increased breathing rate (BR) and higher-frequency ventilation (?).

The raw ECG signal was first band-pass filtered between 8 and 20 Hz using a Butterworth filter, as this passband has been shown to be optimal for QRS complex detection with a favourable signal-to-noise ratio (?). R-peaks were then detected in the cleaned signal using the QRS detection algorithm proposed by ?. Subsequently, the signal-quality index (SQI) mechanism from ? was applied to identify low-quality ECG segments, and HR was computed from the detected R-peaks only for 25 subjects with ECG recordings classified as having excellent quality.

For the RES signals, an optimised breath detection algorithm developed by ? was used to preprocess the respiratory traces and to derive breathing rate and respiratory peaks. This method has been shown to outperform conventional algorithms on breath detection in terms of receiver–operating–characteristic performance. Respiratory amplitude was computed as the difference between each respiratory peak and its preceding trough.

Because ECG and RESP were recorded using the same sensing device, no additional timestamp alignment was required between these two signals. For each subject, difficulty level, and run, physiological and aircraft features were extracted within a sliding 10 s window that was aligned to the performance timestamps. First, the performance, ECG, RESP, and aircraft data streams were converted from MATLAB time format to seconds and aligned by correcting for the offset between the start times of the performance and physiological recordings, yielding a common time axis for all signals. A 10 s window ($t - 10, t]$ was then defined and advanced in 1 s steps along the aligned performance timeline. For each window, the last available performance sample was used to assign the glideslope, localizer, airspeed, and total error as target variables at time t .

Within the same 10 s interval, the corresponding segments of the cleaned HR and respiration signals were selected, and the following time-domain features were computed: the mean, standard deviation, minimum, maximum, and range of HR; the mean and standard deviation of BR; and the mean and standard deviation of respiratory amplitude. In parallel, the mean aircraft elevation trim, aileron trim, and landing-gear position were computed over the window. Each window thus yielded a single feature vector, comprising physiological and aircraft features, paired with the contemporaneous performance errors, resulting in a second-by-second sequence of 10 s summary observations for subsequent modeling.

4.3 Machine Learning Setup

All predictive analyses were conducted on subject-level dataframes constructed from the 10-s windows described above. For each subject, difficulty level, and run, physiological, aircraft, and performance information were aggregated into a feature matrix and a multi-output target matrix, where each row represents one time window and the targets correspond to glideslope, localizer, airspeed, and total error. Two feature configurations were considered. In the no-lag condition, the feature set contained only physiological and aircraft features derived from the current window. In the lag condition, the feature set was augmented with lagged performance measures corresponding to the four target variables at the preceding one and two seconds, thereby enabling the models to exploit very recent performance history.

To quantify generalization, two cross-validation strategies were implemented that respect the hierarchical and temporal structure of the data. For subject-level generalization, leave-one-subject-out cross-validation was applied using the subject identifier as grouping variable, such that all windows from one participant were held out as test data in each fold and the models were trained on the remaining participants. For within-run temporal generalization, for each subject and run the final 60 s of data were excluded from the training-validation set, flagged as temporal test segments, and retained exclusively for final evaluation. On the remaining data, grouped 5-fold cross-validation was used, with groups defined as unique subject-run combinations so that all windows from a given run were assigned to the same fold. In both evaluation schemes, hyperparameter tuning was carried out within the respective cross-validation procedure on the training-validation set only; the selected configuration was then refitted on the full training-validation data before testing.

Three tree-based regression algorithms were used for model training and evaluation: Random Forest (RF), XGBoost (XGB), and CatBoost (CB). Barry et al. (2025) implemented the same models on this dataset using an ad-hoc approach to predict performance error; the present study extends this work by systematically examining real-time forecasting performance across feature configurations (lag vs. no-lag) and generalization regimes. For each algorithm, a discrete hyperparameter grid was specified to reflect key design choices (e.g., number of trees or boosting iterations, maximum tree depth, and column subsampling rate), and hyperparameter optimization was performed using exhaustive grid search with cross-validation (GridSearchCV). The negative root mean squared error, averaged across output variables, served as the selection criterion for the best estimator.

The explored hyperparameters and their candidate values are summarized in Table 1.

Table 1: Candidate model hyperparameters

Model	Hyperparameter	Values
Random Forest	Number of Estimators	[50, 100, 150]
	Minimum Samples per Split	[2, 3, 4]
	Minimum Samples per Leaf	[2, 3, 4]
	Bootstrapping	[True, False]
XGBoost	Number of Estimators	[10, 100]
	Maximum Depth	[2, 3, 4, 5]
	Columns Sampled by Tree	[0.3, 0.4, 0.5]
CatBoost	Maximum Depth	[2, 3]
	Learning Rate	[0.01, 0.02, 0.03, 0.05]
	L2 Regularization Coefficient	[6, 10]

4.4 Evaluation Metrics

Model performance was evaluated using root mean squared error (RMSE), mean absolute error (MAE), and the coefficient of determination R^2 . RMSE quantifies prediction error in the same units as the target variables and places greater weight on larger errors (Barry, 2025), whereas MAE provides a complementary, more robust measure of typical error magnitude that is less sensitive to outliers. R^2 captures the proportion of variance in each target explained by the model and serves as the primary indicator of overall model fit in the Results section. For each cross-validation setting, the mean and standard deviation of these metrics across folds are reported, along with an overall mean aggregated across targets.

Furthermore, to assess whether the models preserved the relationships between specific error components and overall performance, Pearson correlations between component errors (glideslope, localizer, airspeed) and total error were examined for both true values and model predictions. These correlations were computed across all test samples within each evaluation setting and are reported alongside the primary error metrics.

4.5 Implementation

All preprocessing, modeling, and analysis code was implemented in Python using NeuroKit2, pandas, NumPy, and scikit-learn for data handling and the baseline model, XGBoost and Catboost for gradient boosting models, and Matplotlib/seaborn for visualization. Model explainability was ex-

amined using SHAP (SHapley Additive exPlanations) to inspect feature contributions for selected outputs. For each final model, SHAP values were computed on a subsample of 1,000 time points, and summary plots as well as local waterfall plots were generated to identify which features most strongly influenced the predicted performance errors.

5 RESULTS

This section reports the performance of three tree-based models for task performance prediction. First, overall subject-level and temporal generalization performance relative to the Random Forest baseline is summarized; next, error correlations and feature importance are examined.

5.1 Subject-level Generalization

In this section, how well the three models generalize to unseen pilots in the subject-level evaluation is presented, first in terms of average performance across subjects and outputs, then in terms of individual error types and per-subject variability.

Table 2 presents the mean MAE, RMSE, and R^2 across outputs and subjects for the Random Forest baseline, XGBoost, and CatBoost, with and without lag features. In the no-lag condition, all models generate negative overall R^2 values, indicating performance below a simple mean predictor, with Random Forest showing the worst fit and CatBoost the best of the tree. When lag features are included, overall R^2 increases to approximately 0.96 for Random Forest and XGBoost and to 0.97 for CatBoost, accompanied by marked reductions in MAE and RMSE.

Table 2: Overall subject-level generalization performance of the models, with and without lag features

Model	Mean MAE	Mean RMSE	Mean R^2
Random Forest (no-lag)	0.71	0.97	-1.21
Random Forest (lag)	0.07	0.12	0.96
XGBoost (no-lag)	0.69	0.91	-0.83
XGBoost (lag)	0.07	0.12	0.96
CatBoost (no-lag)	0.67	0.89	-0.67
CatBoost (lag)	0.05	0.10	0.97

Per-output performance for the best subject-level model (CatBoost with lag features) is reported in Table 3. For glideslope and localizer

errors, R^2 values of approximately 0.98 indicate that the model captures most of the between-subject variance in these errors. Airspeed error is slightly more difficult to predict, with R^2 values around 0.96, whereas total error shows somewhat lower but still high explained variance ($R^2 \approx 0.97$). The corresponding Figures 1–4 show that predicted values cluster tightly around the identity line for all outputs, confirming that both the direction and the magnitude of subject-level deviations are reproduced well.

Table 3: Per-output performance for CatBoost with lag features

Output	MAE	RMSE	R^2
Glideslope error (deg)	0.03±0.01	0.07±0.05	0.98±0.03
Localizer error (deg)	0.03±0.02	0.08±0.09	0.98±0.02
Airspeed error (kts)	0.07±0.01	0.12±0.02	0.96±0.04
Total error	0.08±0.03	0.14±0.04	0.97±0.03

Per-output performance for the best subject-level model (CatBoost with lag features) is reported in Table 3. For glideslope and localizer errors, R^2 values of approximately 0.98 indicate that the model captures most of the between-subject variance in these errors. Airspeed error is slightly more difficult to predict, with R^2 values around 0.96, whereas total error shows somewhat lower but still high explained variance ($R^2 \approx 0.97$). The corresponding Figure 1 shows that predicted values cluster tightly around the identity line for all outputs, confirming that both the direction and the magnitude of subject-level deviations are reproduced well.

Figure 2 compares the RMSE of total error per subject for the Random Forest and CatBoost models with lag features. For all subjects, CatBoost generally achieves slightly lower RMSE than Random Forest. The most noticeable gains occur for subjects exhibiting relatively high baseline errors. For most pilots, changing the model architecture offers only limited additional benefit beyond the inclusion of lagged features. For glideslope and localizer errors, the mean RMSEs in the subject-level setting are low (approximately 0.07–0.08), but the corresponding standard deviations are relatively large (approximately 0.05–0.09). This pattern reflects substantial variability across pilots: some subjects show very small errors, whereas others retain noticeably higher prediction errors, indicating that generalization quality is not uniform across individuals.

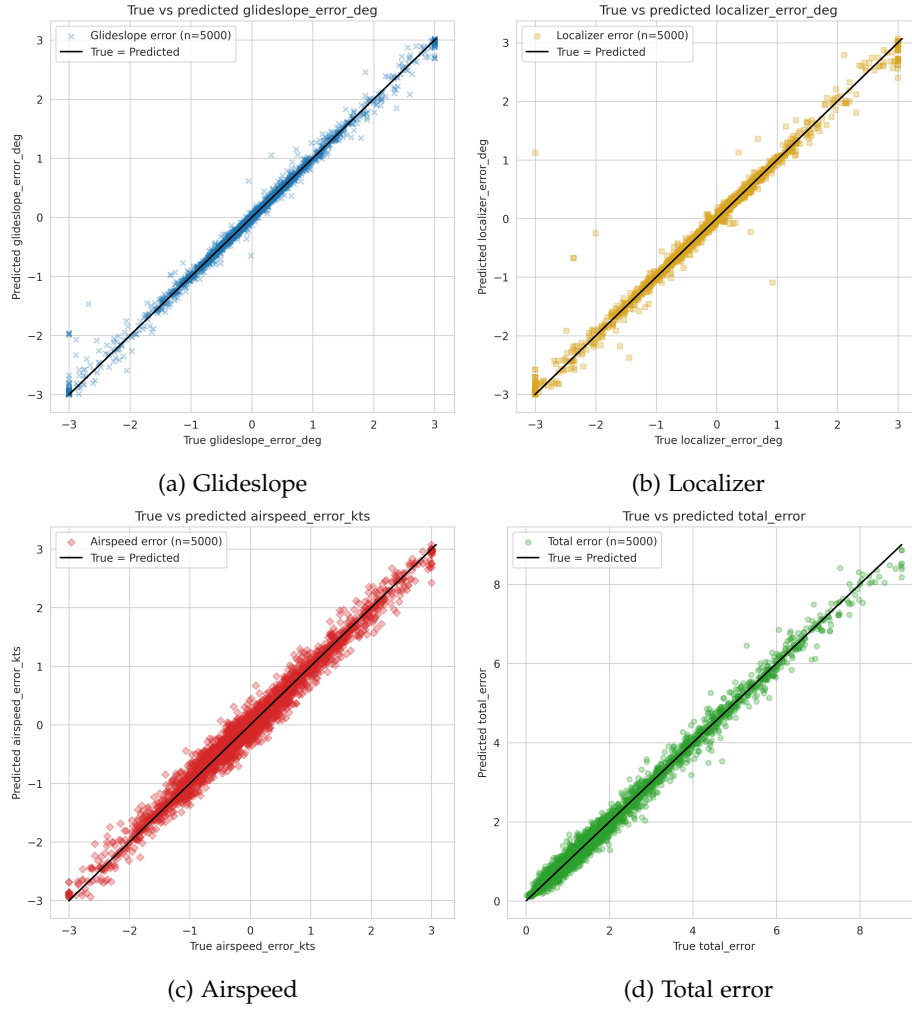


Figure 1: True-versus-predicted errors for the best subject-level model.

5.2 Temporal Generalization

5.3 Temporal

6 DISCUSSION

Here you interpret your findings in light of your research goals and the broader literature. Begin by briefly restating your research goal and summarizing your main findings. Then, discuss how these results relate to the existing research context established in your Introduction, i.e., do they confirm, contradict, or extend prior work? What is the specific contribution of your thesis?

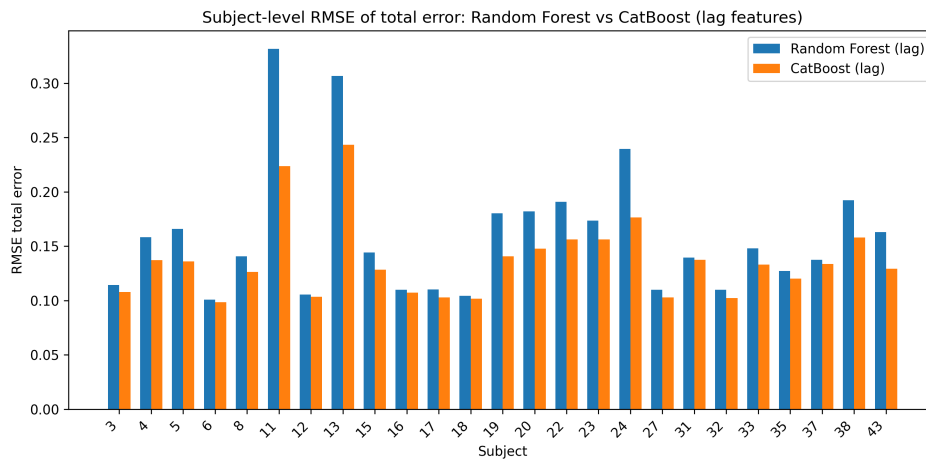


Figure 2: Comparison of total-error RMSE per subject for Random Forest and CatBoost models with lagged features

Discuss your findings in the same order presented in the Results section. Offer interpretations for surprising or unexpected outcomes relying on the existing literature. Acknowledge any limitations of your methods, models, or data that may have affected your results, and explain how these limitations impact the validity and scope of your findings. Where relevant, suggest alternative approaches that could address these limitations.

Conclude by identifying directions for future research based on your findings, what questions remain unanswered? What new questions has your thesis raised?

7 CONCLUSION

Here you provide a concise summary of the overall work in your thesis. Essentially, you want to answer: what are the key takeaways? Clearly articulate the implications of your work: what does it mean for the field, for practice, or for theory?

Keep this section focused and avoid introducing new information, as that belongs in the Discussion.

REFERENCES

- Ananny, M., & Crawford, K. (2018). Seeing without knowing: Limitations of the transparency ideal and its application to algorithmic accountability. *New Media & Society*, 20(3), 973–989.

- MacKay, D. J., & Mac Kay, D. J. (2003). *Information theory, inference and learning algorithms*. Cambridge university press.
- Minsky, M. (1961). Steps toward artificial intelligence. *Proceedings of the IRE*, 49(1), 8–30.

APPENDIX A

Table 4: Multimodal physiological data

Variable Name	Unit	Description
EMG Wrist Flexor	mV	EMG of wrist flexor muscles
EMG Wrist Extensor	mV	EMG of wrist extensor muscles
PPG Finger	mV	Optical signal at tip of the middle finger
EDA Hand L	KOhms	EDA measured on the left hand
ECG Projection LL-RA	mV	ECG Signal: LL-RA projection
ECG Projection LA-RA	mV	ECG Signal: LA-RA projection
ECG Projection VX-RL	mV	ECG Signal: VX-RL projection
Respiration Trace	mV	Respiration Trace
Accelerometry Forearm x/y/z	m/s ²	Accelerometry of the right forearm
Accelerometry Torso x/y/z	m/s ²	Accelerometry of the torso
Gaze Origin L/R x/y/z	mm	Origin point of the gaze ray from each eye
Gaze Direction L/R x/y/z	mm	The normalized gaze direction of each eye
Eye Openness L/R	-	A value representing how open each eye is
Pupil Position L/R x/y/z	-	Normalized position of each pupil in sensor area
Pupil Diameter L/R	mm	Diameter of each pupil

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APPENDIX B

And this!